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Review

Different sensor based intelligent spraying systems in Agriculture

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ABSTRACT

Chemical application of nutrients and pesticides is one of the most important processes in agricultural production, but also one of the most dangerous agricultural operations. To improve the chemical efficacy, reduce chemical and labor costs, minimize labor hazards, and reduce the harmful environmental damage. Variable-rate spray applications that use intelligent control systems can significantly reduce pesticide use and off-target environmental pollution. Real-time variable-rate spraying technology offers effective and efficient use of pesticides. The variable-rate spray allows the farmers to apply pesticides only on the target, using only the correct amount based on the canopy size, season, and growth phase of the plants. In the past few decades, target detection systems have been developed using advanced methods such as laser and vision scanning systems or, more simply, ultrasound, infrared, and spectrum systems. Real-time target detection spray Systems used for the detection of the geometric properties of tree plants are reviewed in detail. Among these methods, machine vision and laser scanners systems are possibly the most capable and complementary means of obtaining three-dimensional images and maps of plants and canopies. This paper discusses a review of various sensing technologies available for the determination of canopy structural parameters and discusses how they are used for precision spraying. Some of the challenges and considerations of the use of these sensors and technologies are also discussed.

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1. Introduction

Chemical pesticides application is one of the essential processes in agricultural production, but also one of the most hazardous agricultural operations. Chemical overflow related with over-application can cause severe environmental concerns. Spraying has a significant role to play to reduce harvest losses and improving productivity. As stated by Cho *et al.* [1] between 30 and 35% of production damages can be saved while dangerous insects and diseases are reduced by spraying. Plant protection is essential to guarantee a good quantity and quality of the yield. Pesticides have become an essential contribution to agricultural production; for that reason, effective use of pesticides is important. Excessive use of pesticides can cause various negative consequences such as pest's growth of immunity to strong pesticides, the destruction of many beneficial pests, and natural enemies of pests, together with harmful pests, causing an increase in populations of insects. Extreme use of pesticides makes it impossible for plants pollinated by insects to bear fruit. The method of applying pesticides shows a significant part in targeting the exact quantity and quality of the spray to reduce environmental contamination and increase sustainability [2]. Spraying with a site-specific application rate will help decrease the quantity of pesticides used in precision agriculture and precision fruit cultivation. Only 30%–40% of the pesticides deposited on the targets. And most of the pesticides lost in the surroundings, causing contamination to the workers and severe environmental pollution. And the purpose of protecting plants with different chemicals is to prevent plants from infection [3].

The drift of spraying may cause damage to non-target plants, contaminate watercourses. Spray drift from air-assisted variable sprayers is a severe and expensive problem for farmers [4,5]. Spray drift is defined as the movement of airborne spray droplets beyond the projected area of the application produced from aerial or ground-based spraying methods [6]. It's found that a requisite amount of the pesticide misplaced in the air and ground [7,8]. And when pesticides are applied, up to 30–50 % of the applied quantity

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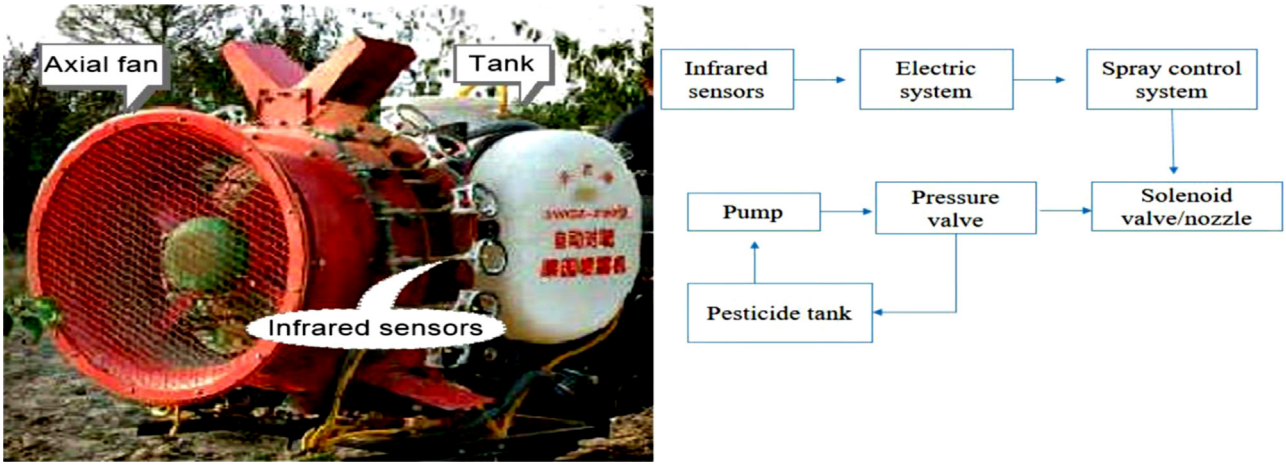

Fig. 1. Automatic target detecting orchard sprayers and System block diagram [33].

of pesticides can be lost in the air, water and other several ways [9]. Therefore, trends in the modern world highlight the importance of the use of sensors and precision agriculture for the effective use of pesticides. Canopies are variable in space and knowledge of the physical features of the tree canopy is an important consideration for the effectiveness of the spray application method for plants. In this study, it is used for the recognition and geometric classification of a canopy for adequate pesticide protection of plants Variable rate spraying allows farmers to apply pesticides only to the canopy target. The overview of electronic methods helps the progress of innovative equipment decrease operational and environmental expenses by improving the effectiveness of the pesticide treatment. The use of a sensor to track the canopy, distances, and space ensures better use of expensive inputs, resulting in a sustainable method to essential agricultural practice. The characteristics of the canopy trees in the garden, the volume of a row of trees, is a very complex study in which the TRV (tree row volume) model is suitable to describe the geometric structure of the plant canopies [10]. An essential objective of the spraying methods is to adjust the practical limitations according to the target canopy density in real-time and to improve the deposition and reduce its flow. The sensor-based variable rate target spaying system has developed for cabbage crops that only spray in the existence of foliage, not in between gaps [11]. Nowadays, it is a growing interest in spray technology to decrease contamination throughout spraying processes. In spray applications, real-time data on the sprayed object can be obtained to control the actual pesticide spray application at variable flow rates. As variable rate technology (VRT) has the potential to make more efficient use of inputs, increase crop productivity and prevent environmental pollution from environmental degradation [12]. Current technology can reduce the use of pesticides and unwanted losses, and therefore their usage is beneficial for farmers, consumers, and the environment. Sensors can obtain the target parameters of the tree and the calculated amount of pesticides can be applied through the nozzle with the help of a suitable data processing unit.

2. Real-time sensor systems

The design of sensor systems to detect target and geometrical structure of plant canopies has discovered in past few decades. The sensors transform the measured physical dimensions into analog electrical data. The working principle of the sensor systems is based on contact with the environment, and the feedback is converted into an output signal and is organized by the technical procedure. Numerous physical phenomena can be useful during the production of a sensor. The dimension of non-electrical signals starts with the conversion into electrical signals, which are then processed. It is essential to identify that each sensor has its limits and small dimension precision, which may differ with changed field situations. Efficiency and accuracy increase with the progress of sensational methods in agriculture [13]. The main characteristics of the sensor is provided in Table 1.

Table 1
Key properties of different sensors [14–16,26,53,54].

Key properties	LIDAR	Ultrasonic	Machine Vision	Infrared
Working range: < 2 m	–	√	√	√
Working range: >50 m	√	X	–	√
Angle measurement: < 10°	√	–	√	√
Angular resolving power	√	–	√	√
Direct Velocity measurements	X	–	–	X
Functioning in rainfall	–	–	–	–
Functioning in Fog or Snow	–	√	–	–
Functioning in Dust	–	√	X	X
Functioning in night	X	N.a.	–	√

X Not possible; – possible, but limitations; √ good performance; N.a. not applicable.

3. Basic principle and characteristics of intelligent variable rate spray technology

3.1. Basic Principle of variable rate spray technology

The sensor is used in variable rate spray technology to detect the target and its geometrical structural features as well as data collection and processing for specific targets. The control unit adjusts the spraying rate of the spraying device in real-time, and the spraying amount is adjusted in real-time according to the geometrical structure of the plant. Due to changes in the entire canopy and the density of leaf, all parts of the canopy require precise spraying [17].

3.2. Characteristics of variable rate spray technology

Variable-rate real-time spray technology is extensively used for pesticide spraying in agriculture. Advanced countries give countless importance to the investigation and progress of spraying systems at various levels. The investment is significant and a more developed, therefore variable rate pesticide spray application system based on real-time detection technology is emerging faster in the last few decades. There is currently international cooperation on the potential of variable rate spray technology and the reduction of pesticide deposits and ecological hazards. Also, the variable-rate sprayers can be adjusted in real-time according to the variations of

the influencing factors [18–21]. The following are characteristics of variable rate spray technology:

- Maximize the application rate: It is possible to achieve a full spray measurement, semi spray, and without spray depending on the thickness of the leaves and it can also make effective penetration, adjust the initial spray speed and improve the deposition of pesticides. Reduce the fluid pressure generated by the spraying processes.
- Saving pesticide: the information received from the sensor, investigates the target structure, and then calculates the quantity of pesticide. This method can reduce the harmful effects of excess pesticides.
- Real-time: For the accurate application of pesticides, variable-rate spraying can be timely and precisely adjusted by parameter changes.
- Cost-saving: Variable-rate spraying can obtain the benefits of pesticides. Significantly decreases the quantity of pesticides, saving labor and physical resources, reflecting clear profitable.
- Higher compatibility: According to the appearances and differences of the plants. Depending on the spraying needs, most suitable detection device modules can be collected or analyzed.

Real-time Sensors and control systems for sprayers are required to obtain a constant deposition of spray in canopies and to reduce spray loss. The objective of this study is to review different real-time variable rate sensor-based spray technologies and the efforts of several researchers utilizing variable-rate pesticide spraying in agriculture.

4. Infrared sensor system

The temperature of all objects higher than absolute zero releases heats energy in the form of radiation. The infrared sensor transmits and receives a high-intensity light pulse to detect the presence of objects in the range of the sensor. In general, measuring distance using infrared detection requires more sophisticated methods, such as measuring phase transition or triangulation [22]. The use of an infrared sensor is an automatic object recognition system. Infrared device detection methods for target recognition and automatic control of the spray system have been adopted in orchard sprayers with automatic target detection. These spraying systems can be easily promoted due to the low cost of detectors with infrared sensors. Infrared sensor has low prices and quicker response times than ultrasonic sensors (US). The indoor performance of the infrared sensor is better than outdoor. The infrared sensor as well depends on the type of objects to be detected, regardless of whether the object colored is light or dark. It's also observed in [23] that in direct sunlight, the sensors were able to detect twice as many solid light-colored objects at a distance of approximately 25 cm. But for dark objects, these sensors are invisible, especially outside. Many developed countries such as the United States, the European Union and Russia are developing automatic sprayers for target detection using infrared imaging techniques [24–26]. The recognition of infrared sensors was studied from the beginning to the present day under two main aspects: number one, the model method. First, place a model, then combine the image to identify with the model and finally perform target recognition, characteristic method number second. The analysis of the whole image becomes a study of specific physical characteristics of the image [27–29]. Which considerably decreases the calculation volume to detect an infrared image, the target area is cleared, so identification is based on the function with the improvement of artificial intelligence, and it has many advantages [30–32].

He et al. [33] designed an automatic infrared target detecting electrostatics, and air-assisted orchard sprayer. The techniques of automatic target detecting, electrostatics, and air-assisted spraying were presented in this system. The sensors hit the upper, central and lower parts of the canopy to detect the shapes of the different fruit trees and signal the control system. Experiments show that the new target detection sprayer with the automatic infrared sensor can store more than 50–75 % of pesticides, increase the utilization rate (over 55 %), increase the sensor effectiveness and reduce the pollution caused by the application of pesticides (Fig. 1).

Another automatic sprayer with two infrared sensors mounted on each side of the machine, respectively, are used to identify sprayed fruit plants. The control system processes the pulse signal from the Hall sensor and determines the tractor speed in real-time. If the target is set, the condition and speed can analyze the position and width of the spray. After the tractor has travelled a certain distance, the spray valves are activated. If the target is detected, the sprayer starts spraying, and if there is no object detected, the sprayer stops spraying. The sensor (<=70 cm) is able to detect the canopy of trees accurately. If the target is detected, the sprayer will spray pesticides on the target within a certain width [34]. Jianjun et al. [35] reported an infrared target detection system containing a combined circuit for an automatic orchard target sprayer. The system met the specific design necessities for strength, sensitivity, dense volume and interfering with environmental radiation and the distance detected between the sensor and target.

Bargen et al. [36] developed a near-infrared reflection detection system to detect plants. These reflection characteristics were determined using spectral radiometry technology. The identification of the plant is based on the different features of the plants. Optical filters were used to select the sensitivity of the spectral bandwidth of red radiation and near-infrared detectors. In the sensor view, the inverse values are digitized to include a standard difference index to provide a more reliable signal of an existing living plant. This sensor system is integrated into a microcontroller to activate a solenoid spray nozzle on a prototype agricultural sprayer. The system can detect the presence of a plant in the area of approximately 20 cm^2. During infrared image processing the next step is feature extraction and it will have impact on target recognition [37]. The visual characteristics of the image are obtained by processing the infrared image. This makes the target and rearview more independent and significantly reduces the degree of interaction and makes the field of interest more visible. To prepare for the next step in infrared imaging, you need to select a function that represents the target. The feature extracted data is used as input for target detection during spraying process [38]. Infrared sensors use the reflected light intensity to calculate the distance of an object are not very common, and only a few researchers are reported in the literature [39–42]. Still, it's essentially quick reaction very interesting for increasing the real-time response of a mobile robot. Also, high-end converters are easily accessible to provide improved angular resolution than American routers. Therefore, it is valuable to explore the progress of low-cost IR sensors that can accurately measure distance with the minimal response time. Some IR sensors reported low cost built on the intensity of the reflected light.

Infrared target detection technology in the targeting of tree plants is more appropriate for large and dense objects reflectors with strong light intensity. It gives the maximum sensitivity of the detector near the middle of target detection distance and best detection outcomes for plants with significant leaf reflectivity. Although these sensors generally do not return the actual distance to an object, they do mean whether something is present in the detection cone or not. These sensors systems are sensitive to external noise. The infrared light existing in ordinary light effects with the real reflected infrared signal. The infrared sensor performance is better indoors than outdoors. The infrared sensor also depends

on the type of detecting objects, for example, if the object is light or dark in color. However, when infrared target detection is used for the method of spraying plant pesticides. The infrared detection system of the automatic orchard sprayer was challenging to operate well and interfere with the severe environment resulting from design errors, including those that can be detected in a short distance. The cost of the automatic target detector is high. Though temperature and humidity have little effect on detection results but the appearance of plants, light intensity, the travelling speed and area of vegetation have impact on detection results, especially the presence of plants and the intensity of light. The density of vegetation and intensity of light is proportional to the detection object distance and width. The detector can effectively differentiates the minimum distance of a single plant. The space of the plant is monotonously correlated to detect sensitivity [43]. Moreover, because of the limitation of this sensor, the target detection technique based on infrared technology is unable to identify typical data like the detailed dimensions and size of the object, the calculation and analysis of quality cannot be implemented. Furthermore, the target detection method is easily exposed to external light [44,45] and modern agriculture cannot meet the needs of the progressive development of technology with the growing demand for the spraying operation.

5. Ultrasonic sensor system

Ultrasonic sensors based systems can measure distances very quickly and automatically. Ultrasonic sensors measure time delay information to measure distance. This delay time between transmission and the reflection of the sound impulse is proportional to the distance between the sensor and the object. And the speed of Sound is used to calculate distance [46]. The ultrasonic sensor consists mainly of two main parts, the transmitter, and the receiver. The transmitter sends an ultrasonic pulse to the target object and the reflected pulse receives at the receiver. The time difference between the two pulses indicates the flight time. And the target distance is calculated by multiplying the flight time by the speed of sound. It's reported that ultrasonic sensors are highly valued for their sensitivity range and can measure canopies of plants. And the vehicle travels speed was 1.8 m/s with high precision. And also reported the requirement of the sensor is that the target should be perpendicular to the sensor for a precise reflection of the waves [47,48]. But, when used in gardens, the environment is not established and therefore it is not possible to predict such surfaces. Consequently, it has limited use. The possible application of the ultrasonic sensor consists of orchard based management fast tree volume. The information can be applied to the various use of agrochemicals in a farm. There was no spraying in the absence of plants, semi-spray when the vegetation is in small size in front of the sensors and it is completely sprayed when the sensors sensed the thickness of the canopy above a certain level. This result has led one way to continuously change the flow due to the variability of the canopy in rows of citrus fruits, vines and orchards [49–51].

Researchers have designed several variable rate spraying systems with ultrasonic sensors. Giles et al. [52–54] a conventional air-blast orchard sprayer by integrating ultrasonic sensor technology into a sprayer control system to measure foliage volume and then control the spray output. Spray savings ranging from 28 % to 35 % and 36%–52% were reported in peaches and apples, respectively. Also, compared with a standard sprayer, the sensor-controlled sprayer had a significant reduction in off-target loss. Later an experimental electronic control system was developed and was able to change the application rate based on tree canopy width [44]. The ultrasonic sensors can measure the distance between the sensor and the outside edge of the canopy and the canopy width

Fig. 2. Ultrasonic sensors based variable-rate sprayer for nursery inside layers [122].

at the same height as the ultrasonic sensors. With known values of tree row spacing, sensor location on the tractor and travel speed, the optimal spray volume was planned and was used to trigger solenoid-valve for control of the nozzle flow rate. Liquid savings of 70 %, 28 % and 39 % were reported in the pear, olive and apple coppices respectively from the variable rate sprayer compared to a conventional application. Poorer spray deposition on the canopy but a higher percentage between the entire spray deposit and the liquid spray output (i.e., better Application efficiency) were also reported. Using the same sprayer Llorens et al. [55] compared conventional spray application and variable rate application in three vine varieties at different crop growth stages. The variable rate application was reported to have an average of 58 % savings in application volume with similar or even better leaf deposition.

Several studies have been conducted to measure the size of the canopy in gardens automatically. Ultrasonic sensors have been used for various purposes in agriculture for decades [56–58,122] as shown in Fig. 2. One of the application consists of detecting the trees to obtain canopy structural information. Gil *et al.* [51] modified a multi-nozzle air-assisted spraying system with ultrasonic sensors and three solenoid-valves. The nozzle flow rate was modulated in real-time as a function of crop width in a vineyard, as measured with the ultrasonic sensors. The canopy in the vineyard was divided into three height sections, each covered by one ultrasonic sensor and one solenoid-valve controlled nozzle. A saving of 58 % percent was reported compared with a conventional constant rate application while deposition quality on leaves, uniformity of liquid distribution and capability to reach the inner parts of the crop remained similar.

An automated garden sprayer is capable of selectively spraying pesticides based on the characteristics of the target. Ultrasonic sensors identified the outline of the apple tree cover. The ultrasound signal is accomplished by a particular computer and fed in real-time to be sprayed opening and closing nozzles associated with the canopy structure. Three ultrasound sensors and three nozzles were used. Sensors used in this experiment provided only information in the form of presentation of the target and its distance [59]. In [60] a variable-rate sprayer was developed and validated. The applied volume of pesticides can vary according to structure of the target by using an algorithm based on a canopy volume stimulated by a series of tree volume models. The differences of the canopy width along the plant line are measured electronically using various ultrasonic sensors located on the sprayer and used to change the flow rate from the nozzle point. The purpose of this method is to maintain a uniform volume of spray coverage. Field tests in different crop phases showed a good relationship between volume

Fig. 3. Variable-rate sprayer with the ultrasonic sensor and component on the sprayer (Reprinted with permission from Ref. [60]. Copyright© 2013, Elsevier).

and canopy properties. Possible pesticide deposits have been estimated at 21.9 % compared to the cost of a conventional application. This result is consistent outcomes of related studies in automated spraying systems.

Zaman and Salyani [61] assessed the repeatability of ultrasonic tree volume measurements, determined the effect of spryer velocity and plant volume density on ultrasonic frequencies and measured the variance between the citrus leaves on the north and south cover. And study the impact of canopy density and ground velocity on the performance of an ultrasonic system that measures the volume of trees. The density of canopy leaves had a significant effect on ultrasound measurements of the size of the canopy. The volume difference was more significant in light trees than in dense trees. The forward speed (1.6–4.7 km/h) had no significant effect on the measurement of the ultrasonic volume. In partially defoliated canopies, the variability of measurements increases with increasing ground speed. There was a substantial difference between the volumes on both sides of the trees canopies (Fig. 3).

Schumann et al. [56] present a real-time ultrasonic mapping program of the tree canopy. A Diagram layout of a manually-measured tree dimensions and ultrasound transducer system was used to calculate the dimensions of the forest cover in a citrus orchard. Vehicles and trailers with a perpendicular network of 10 ultrasonic transmitters and a comprehensive differential positioning system (DGPS) measure tree heights. The converters can be installed from 0.6 up to 6.0 m from the ground surface (Fig. 4). The information collected by an automated system compared to manually-measured measurement data from 30 trees to evaluate precision 376 citrus fruits were examined twice to evaluate repetition. The results have been shown there is no significant difference between the dimensions of the ultrasonic and manually measured height. The system also located the tree structure location points for GIS mapping.

Another author [58] recommends a real-time technique founded on a series of ultrasonic sensors to estimate the coverage density. This evaluation can be used as a reference to adjust the parameters of the bed spray machine to improve spray deposition and prevent drift. Two experiments were conducted, the first to determine the signal behavior with an ultrasonic sensor in a greenhouse and to adjust the algorithms. The second series of tests were conducted in real working conditions in the garden. The results show that the signal obtained is connected to the growth period and has the same values on both sides of the rows, with an error of 14.1%, 3.8% for vines and apple trees respectively, and it is susceptible to fog detection. The canopy echoes and density are proportional.

An essential goal of spraying systems is to adjust the operating parameters in real-time according to the target density to keep the droplets deposition under the canopy, so increasing the deposition of the sprays and reducing the drift of the sprays. An apple orchard and two vineyards were analyzed using an ultrasound scanner per week and the sensory data were correlated with the analysis data in the fourth point. The results showed that the system had to be calibrated for each plant or variety and that the ultrasound system could detect density with a 4.76 % error in total coverage in the first period, mid-season and before the harvest date [62].

Ultrasonic sensors are often used for calculation of the volume of the canopy of tree plants and to adjust the spray volume. Various laboratory tests have been performed to verify the suitability of a single gain broad spectrum ultrasonic sensor, such as the experimental determination of the sound cone diameter at different distances for different target materials and the determination of the sound effect of the angle wave propagation. Determine the magnitude of errors and the target distance due to the accuracy of the measurements for different materials and the interference between the sensors for two different materials [63]. The results show that the sensor is suitable for olive trees since the sound cone is thinner for an olive tree than the other elements. The coverage of the olives does not affect the accuracy of the sensor due to the distance and angle, meddling errors is negligible for large sensor deviations and sensor distance measurements were considered entirely accurate. Similarly the crop identification system (CIS) by using ultrasonic sensors has developed to determine the characteristics, dimensions, and density of the target. Sprayer prototype can automatically adapt the sprayer and the air distribution according to the features of the target canopy, the level of disease of the crop or environmental conditions. The operation of the CIS system and the first experimental results achieved from the deposition tests carried out in an orchard using a prototype sprayer

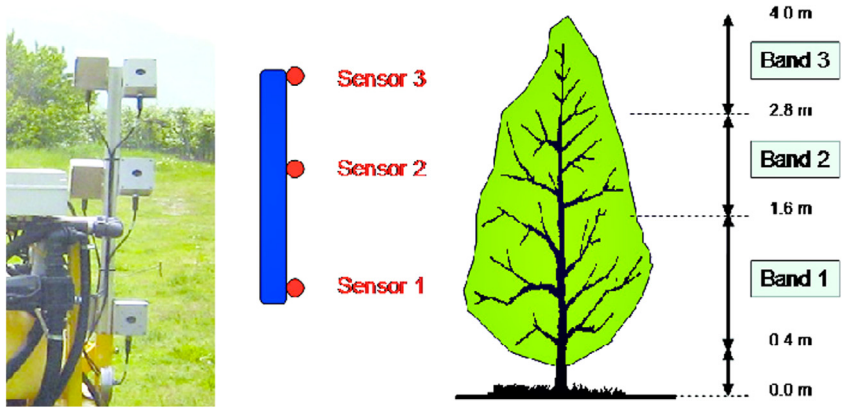

Fig. 4. Ultrasonic sensors mounted on the sprayer and schematic diagram of sensor positions [64].

equipped with CIS. The first results showed that, by adjusting the application rate to the features of the target, the CIS allowed to maximize the quantity of spray applied to the leaves in always higher volumes [64]. In [65] to determine the durability and precision, an ultrasonic sensor simulated the environmental and operating conditions. Test conditions include prolonged cold, outside temperature, crosswinds, temperature changes, and movement speed and effects of the spray. The root mean square error (RMS) in the sequence of distance measurements from the replicated vegetation was used to verify a significant difference between treatments. After four months of cold weather outside, the RMS distance error measured by the ultrasound sensor was increased. The physical position of the spray nozzle associated with the ultrasound sensor significantly influenced the mean RMS errors. RMS errors can be minimized to suitable values by adequately checking the space of the spray nozzle. Also, the stability and accuracy of the measurement were tested when many coordinated sensors were installed in the sprayer prototype. Separation of the ultrasonic wave path from each sensor has been found to reduce interfering between sensors during the multisensory process. The test methods are useful in the design of standard test procedures for distance sensors for use in the field. So far, only ultrasonic sensors have been extensively used for developing variable-rate sprayers or orchards, nurseries and vineyards. However, there is a growing number of investigations on using ultrasonic sensors and other types of sensors to measure tree crop canopy structure and characteristics, which is a crucial step in developing a variable-rate sprayer for tree crops. The benefits of using this ultrasonic sensor are its availability, durability, and low price.

Ultrasonic sensors are among the different types of sensors used in present precision spraying methods. It is relatively resistant in favorable conditions, outdoors and can evaluate the canopy. The volume of the trees has been used successfully by many researchers. It was revealed that the ultrasound system could detect density. But it has some advantages and disadvantages. It is not suitable for changing environments because it uses the constant speed of sound to calculate the distance. The speed of sound in the air is a function of temperature and the amplitude of sound pressure decreases as sound gradually loses friction in the transmission medium. Ultrasonic sensors are sensitive to background noise; as the frequency increases, the ultrasound background noise level decreases. And the resulting noise increases as it passes through the air. However, the increase in the frequency of the sound waves is limited by the unwanted side effects of an increase in the propagation of energy [66]. Also the specular reflection can occurs when the ultrasonic beam cannot return directly to the receiver because it is far from the target object [67]. This means that no obstacles have been found for a mobile robot. Group of researchers [26,53,54,68,69] studied in various aspects of the applications and compared the laser and ultrasonic transmitters for limiting plants and measuring citrus trees. Laser sensors work better than ultrasound due to their higher resolution.

6. LIDAR sensor system

LIDAR stand for Light Detection and Ranging. It is a remote sensing technology which uses the pulse from a laser to collect measurements which can then be used to create 3D models and maps of objects and environments. Laser LIDAR is an active sensor; emits a temporarily modulated laser radiation to a target that reflects radiation to an optical receiver located at a point. LIDAR calculates the distance by the flight time of the pulsed light. It's the time when the light is sent and received. A very short pulse (laser beam) is transferred to an object. Then part of the light is reflected in the section, and the second part is reflected later. A rotating glass directs a pulsed light in many points of the semicircle. An angular sensor gives the exact path on the LIDAR mirror. A large number of measured coordinates are combined to create a model of the contours of the surrounding areas. A flat scan provides information on a ladder. This can be removed by turning the laser source to obtain a 3D image. Laser range sensors are actual precise for calculating distances, with a precision of 1 cm higher than a length of 80 m. This makes them suitable for control applications. Laser radars have been used to find and avoid obstacles. It has a higher resolution than an ultrasound sensor, which makes it more precise. However, its performance gets worse with dust and rain. Gil and Llorens [70] discussed the use of a scanning Light Detection and Ranging (LIDAR) system to describe pesticide drift during pesticide spraying. The LIDAR sensor presents an encouraging and easy method to found the possible drift of a precise spray position. In [71] a tractor-mounted LIDAR system use for the measurements of dwarf and semi-dwarf Cox apple trees. Which develops structural parameters of the canopy for pesticide spraying. Also presented the distributions of crop area density which can be used to estimate the deposition distribution of spray. In [72,73] used a laser scanner with an offline processing algorithm to scan the citrus canopy. Based on the scanning data, they calculated the canopy characteristics such as tree height, width, canopy volume, foliage density and tree boundary profile. An artificial target was tested and the results showed an accuracy of 97 % for the length measurement. The density estimation was found to have a reasonable correlation with the visual assessments. Another author [74,75] used a laser scanner to measure canopy volume and to estimate tree foliage area in real-time. The validation tests with pear trees concluded there was a linear relationship between canopy volume and the foliage surface area with a coefficient of correlation (R) of 0.81. However, they also

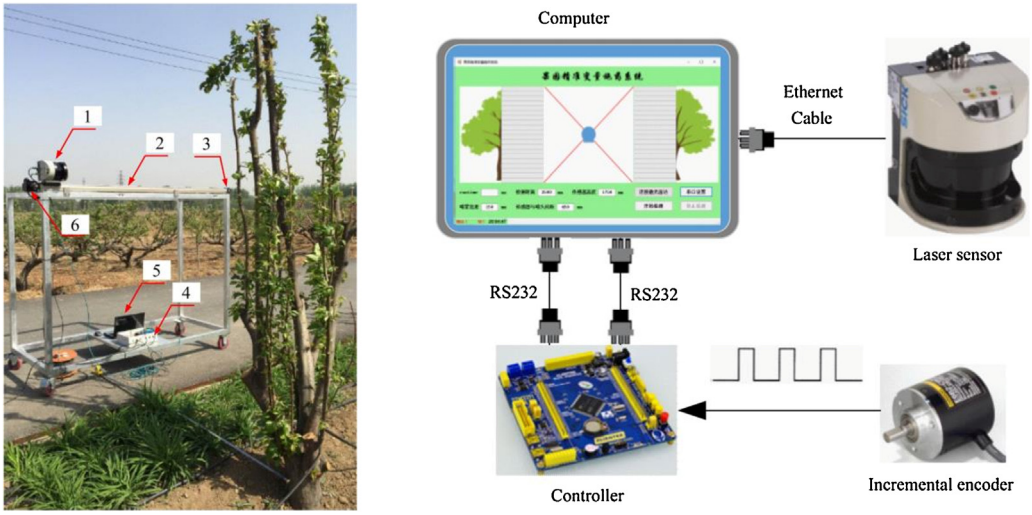

Fig. 5. Structural diagram of LIDAR laser-scanning system [79].

reported that the accuracy of the estimation was closely related to tree species and tree truck size could influence the estimation result. Lee and Ehsani [76] designed a laser scanner to scan citrus trees and develop a program using the saved data to estimate the tree canopy height, width, and volume. The height estimation was compared with manual measurement and the error was found to be less than 1 cm. The width estimation was found to have an error of about 15 cm and its accuracy was believed to be dependent on the accuracy of the tractor speed measurement. The failure of the volume estimation varied with the definition of the profile polygon in the algorithm. A minimum error of 0.09 % was found for the volume estimation.

As demonstrated in [77,78] a laser scanner mounted on a tractor to scan selected trees of a vineyard and a pear orchard several times before and after defoliation. The scanned data was then used to build 3D images to determine geometrical and physical factors of the vegetation such as volume and leaf area of trees. These geometrical and structural parameters were compared with plant leaf apparent values got by labor-intensive measurements in which leaves were picked, and one-sided proposed region of the leaves was measured by the use of shadowgraph techniques. Results have shown a good linear correlation between the canopy volume calculated from the laser measurement and the total foliage area from the manual measurement (Fig. 5).

A LIDAR sensor can calculate the geometric structure of the object. By using a computer the measurement data obtained from a laser scanner and to achieve the range of motion of the laser scan sensor from a controller. Two trees apple and conifer were selected at an adjustable speed system and network size to verify the accuracy of the laser scan. The coefficients of variation of the whole volumes for each tree species were 0.078 and 0.041 respectively, indicating that the laser scanning method is useful in providing penetrating volumes of different densities of the canopies with real-time drive reliability. The laser scanning system delivers a good estimation of the canopy volume in real-time to control the sprayer at different levels as demonstrated in [79].

The LIDAR use in sprayers allows the measurement of essential geometrical parameters of orchards like the size of trees. The possibility of very quickly allows us to calculate the distance between the sensor and the surrounding objects. And obtain 3D cloud points which, the application of suitable algorithms, allow digitization and rebuild and define the tree structure with high accuracy [80]. Even though it has some drawbacks for cloudy environments, but the LIDAR system is one of the best-used sensors for the geometrical description of tree canopies. LIDAR can measure spatial variability, and it is an important feature of plant structure. It is a significant development compared to some previously applied methods. LIDAR systems can be used to measure changes in the canopy structure at different time dimensions, which can deliver complete information on the growth of the canopy and the evaluation of responses to field experiments. The laser system offers unique choices for viewing the angle and distance data needed to model a structure to investigate the LIDAR structure applications thoroughly [81]. The advanced LIDAR system has the potential to obtain 3D digital images with plant information, height, width, volume, leaf area index and leaf area density data. The two-dimensional (2D) surface can be used in agricultural applications. And convenient to use LIDAR sensors [82]. 2D LIDAR sensors obtain a point cloud that corresponds to a plane or part of an object of interest. These sensors alone do not allow you to limit full coverage to 2D perception [83]. Therefore, the whole sensor provides a point cloud that can be used to create a 3D image.

In [84] authors focused on comparing the manual volume of canopies estimations with LIDAR and ultrasonic sensors. The results showed a good correlation between the evaluation carried out by LIDAR and the relationship between the ultrasound sensors and the physical measurements. The observations showed significant differences between physical and sensory assessments, less dense canopies of trees. This canopy data was used to regulate the dosage of agrochemicals [85] and to evaluate the yield of citrus fruit [86]. The LIDAR sensor, combined with a vertical sampling resolution, can collect additional information from the canopy for a more precise estimate than various ultrasonic sensors [87]. These are satisfactory test results, but it is not easy to generalize these results to other trees of different structures. Although some groups have developed models to regulate the flow of the application, despite changes in the structural parameters of the canopy using ultrasonic sensors. A review of several studies on target spraying has shown that the use of variable-rate spraying methods in orchards is still in the prototype phase; however, sprayers are already available to control weeds and fertilize crops [88].

LIDAR has some advantages such as higher accuracy than single-point measurement, for example, when using ultrasonic sensors. Laser LIDAR works very quickly and can be used in all lighting conditions, including experimental light. Laser scanners can collect more target information in less time and are less affected by the air and field environment than ultrasonic sensors. The laser sensor gives satisfactory accuracy and reliability to find out tree structures irre-

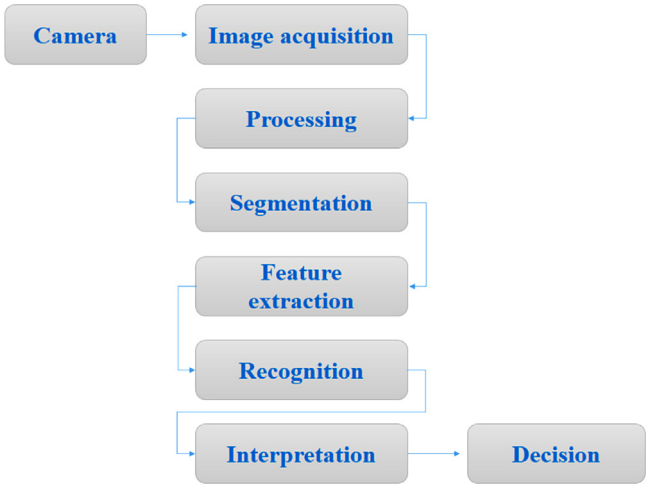

Fig. 6. Computer vision stages.

spective of weather conditions or lighting situations, and they have greater possibilities to be used for the variable-rate sprayer development for orchard and nursery applications than other types of sensors. Laser radars have some limits. Lasers are one of the most common sensors. It sends the laser beam in a straight line so that if something interferes with the laser, it is not useful in this situation. By limiting the sensor, the laser emits more or less a signal that detects specific surfaces. Laser beams cannot enter vegetation and lousy weather, in the snowy or foggy environment, they cannot be used. Based on the previous investigation from other researchers, it can be concluded that a laser scanner can characterize more data on the product structure more accurately than an ultrasound sensor. Lasers offer better resolution and precision than ultrasonic sensors.

7. Machine vision system

Machine vision is an efficient method to provide recognition of vegetation position, size, shape, color, and texture information; the use of computer vision in mechanized agriculture is becoming more and more popular. Machine vision is a computer's ability to s̈ee.Än artificial vision system uses one or more video cameras to acquire images that a computer must interpret. Machine vision is one of the most popular detection technologies for robots. When the road to be crossed is different from the background, automated vision technology can be used to drive an automatic vehicle. Vision management has the benefit of using local functions to control vehicle navigation.

The performance of the machine vision system while driving a car can be compared to that of a high precision laser radar sensor [48]. Machine vision system driving applications include guiding tractors for cultivation and driving an autonomous vehicle in a greenhouse. But, in general, complex methods are needed to extract valuable information. For example, sonar data directly estimate depth; but the vision system requires a compatible stereo algorithm for visual data [89]. For vision technology, the input and output is information about the image. The vision system is divided into several processes [90] as shown in Fig. 6.

Taking a photo is the first step in machine vision using a CCD camera. This operation allows you to separate specific objects or areas after preprocessing the image. The segmented area is analyzed to identify different characteristics. Images are used to identify the identified characteristics.

In most of the applications, only a single value is measured at each sample point of the image, representing the image brightness at that point, this value is called the grayscale, and hence the image is known as the grayscale image. The selective spraying robot proposed by Oberti et al. [91] uses crop image detection to capture Powdery mildew plants, powdery mildew, have powdery spots on the leaves initially, and the powdery mildew symptoms. The white powder gradually became denser and turned to gray-white. The disease spots expanded and became interconnected and covered with whole leaves, which eventually made the leaves yellow. Therefore, the research team used image processing to find the location of the symptoms in the multispectral image. The outcomes show that the robot was capable to mechanically identify and spray pesticides at target unhealthy region within the canopy and to saves the pesticide consumption from 65% to 85% when related to conventional common spraying of the canopy (Fig. 7).

As suggested in [92] a new method for detecting and locating pests based on binocular stereo vision. In this method, the color characteristics between the parasite and the leaves of the plant are removed with an appearance segment to identify the pest. The 3D positioning of the pest was obtained through image segmentation and binocular stereoscopic vision techniques. In the positioning process, the centroid matching technique was adopted to change the compatibility of the common object. Also, a formula based on a binocular stereo vision for distance measurement has been redesigned. An effective way to draw overlapping leaf areas. By the planned process and leaf area approximation technique can minimize the leaf overlap and improve the precision of leaf area assessment and growing rate calculation. Use patterns proposed by Jamil et al. [93] Learning operations to remove noise and enhance the binary image, using six basic morphological operations can effectively remove noise.

In [94] introduced two algorithms to achieve contour extraction and input a binary image. This study uses the contour extraction vision library to find contour in Open CV. Binary image Perform topology analysis. The first algorithm is to determine the surrounding relationship of the binary image boundary and define the outer edge and the whole boundary. The hierarchical relationship between them assumes that the outer boundary corresponds to a connected region with a pixel value of 1 and the entire boundary corresponds to a region with a pixel value of 0. We can use the boundaries to represent the original image. Canny [95] proposed a set of computer vision the multi-level edge detection method is called canny edge detection. This method uses a set of definition methods to define edge point pixels. Research on edge detection techniques for image segmentation also proposed by [96]. The advantages and disadvantages of various methods of edge image segmentation are compared. Segmentation technology, consisting of complementary elements of fuzzy logic, neural computing, and evolutionary computing, improve recognition and recognition capabilities [97].

Using machine vision technology, an orchard sprayer was developed which can apply a specific quantity of pesticides based on a green canopy and increased the efficiency of chemical application in the orchards. In the olive orchard, the sprayer field performance was assessed under continuous and discrete target spraying conditions at three forward speeds and four repetitions. Variable-rate selective spraying can reduce 54% chemical spraying than conventional constant spraying [98]. Another machine vision variable rate sprayer was developed. A monochrome CCD camera was fixed in the front sprayer, the distinction among the weeds of the plant is acquired by processing the image from the spatial data by Gabor filter. This technique lets us identify periodic signals from non-periodic signals and we have a weed patch distribution to increase clipping lines. Therefore, weeds were precisely identified using a blob-coloring technique. To end, by using a pinhole camera to convert the image data of the lawn coordinates into global coordinates to actuate the correct electro-pneumatic coverage of the sprayer [99]. In [100] a sensor system based on machine vision and a selec-

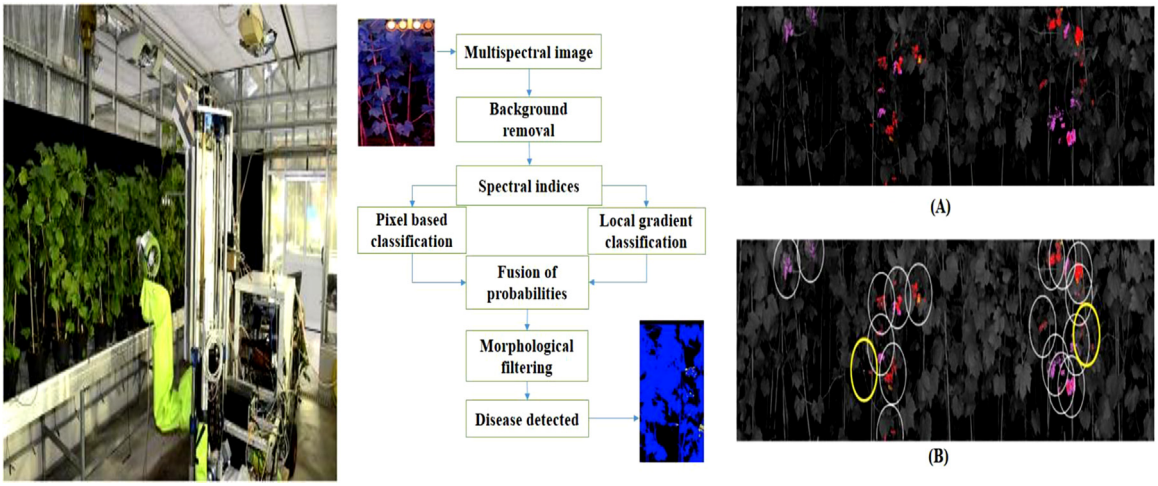

Fig. 7. Targeted sprayer in greenhouse, Disease detection algorithm and spotted disease area. (Reprinted with permission from Ref. [91]. Copyright© 2016, Elsevier).

tive control system of herbicides have been developed and fixed in the sprayer. A Windows-based MS imaging system is integrated with a targeted real-time spray control system to perform remote control operations. A finite state machine (FSM) was used to design the controller, and general design features were developed to determine the displacement distance between locations. Using artificial targets, the accuracy of the spatial application of the system was calculated in the field. It operates at 91 % of the total travel accuracy without statistical evidence that the accuracy of the travel or the average length of the model depends on the speed of the vehicle. The significant variances between the length of the sample and the average width of the model varied from 3.2–14 km / h. Spray pattern tends to flow faster towards the target.

Another author [101] describes the purpose of image processing algorithms with an internal processor capable of detecting pests in agriculture crops. An automatic sprayer capable of spraying pesticides based on the physical characteristics of the pests. And an automatic spray machine with ARM9 processor was experimentally evaluated. Four different agricultural samples were taken and were analyzed using the Moving object Extraction method. It showed the pests appearance with high accuracy of 96 %.

Berenstein et al. [102] proposed plant detection algorithms for a targeted sprayer for vineyards. To reduce the use of pesticides during the fumigation of plants, new artificial vision algorithms have been developed to detect the spaces between the vineyards and determine the exact position of the grape clusters to direct them correctly. By these recognition abilities, a spray robot with a nozzle can selectively and accurately spray, greatly reducing spraying and labor. The results show an accuracy of 90 % in the detection of a cluster of grapes, with a 30 % pesticide use reduction. In this review paper is the one most commonly used Microsoft Kinect technology also discussed. Microsoft Kinect technology provides real-time information on the color and depth of the target. Microsoft Kinect has a red, green and blue (RGB) camera, an infrared transmitter, and an additional monochrome metal oxide (CMOS) semiconductor video camera. In [103] the author modifies and manages the spray concentration and the pesticide dose; this work provides an equation for calculating the average distance of trees. Study results showed that the distances calculated based on the Kinect system data were accurate. And a concept of density to improve the dose of sprayed pesticides. Finally, the results of the peach, apricot and vineyard experiments have shown that a precision pesticide spraying model for smart orchards based on average distance and can increase pesticide spraying efficiency, decrease waste and environmental contamination and manage automatic and precise fruit production.

The reliability of the vision system is reduced by reducing the existence of light, dust, and fog. Benson et al. [104] reported that to overcome this problem by using artificial lights, but faced a problem of vehicle shadow. They were able to reach a speed of 1.3 m/s with a precision of 0.6 cm. Camera placing is also an essential feature in the functioning of the visual method. It is preferable to mount the video camera in the upper front part of the tractor cab, pointing approximately 30° downwards. NIR filters used for images to processing from a CCD camera [105]. The vision system consists of several complex algorithms for processing, processing, and recognizing images. As a result, the amount of computation required increases and therefore special processors are used. However, as computers have evolved, this concern hasn't stopped them from being used for sensing. The color vision system improves the capabilities of the spray system. Color provides an additional source of information for target detection, but it adds the cost of computation. After the recognition phase, the observation phase decides what to do based on the data of the previous stage and the control algorithm. The ability of a computer to collect large amounts of data from a single sensor makes it an excellent choice for autonomous systems. High demand for computation is a severe deficiency of vision systems, which limits their application in real-time. The high cost of the vision system is also a problem in some applications

8. Sensor Fusion system

The data integration and knowledge from various sources are called data fusion. According to Dasarathy [106], "sensor fusion is the study of concepts and techniques developed for optimal information processing in distributed multi-sensor environment through intelligent integration of the multi-sensor data." Multi-sensor environment, by their nature, generates a large volume of different resolutions data. These are frequently corrupted by a variation of noise and clutter conditions and continuously differ because of temporal changes in the environment. Multi-sensor data fusion methods collect data from various sensors, as well as correlated data from related databases, to attain better precisions and more precise implications than could be realized by the use of a single sensor alone. The data from each sensor is different in terms of interpretation algorithm, resolution, the amount of data produced and processing time. In combining this heterogeneous data, sometimes conflicting situations arise [107]. Each sensor provides a separate feature of the environment; the set of sensors can be functional together creating sensor interaction. The nature of these connections determines how fusion is achieved. There are

Table 2
The past decade work on targeted spraying spraying based on real-time sensors technology.

Author	Year	Sensor	Objective of the work
[118]	2008	Ultrasonic	Application of low-volume bait treatments with sensor-controlled sprayer on the exterior of the crop canopy
[119]	2011	Ultrasonic	Development of efficient and accurate variable-rate spraying systems for targeted application of pesticides in blueberry fields
[120]	2012	LIDAR	Improvement of a sprayer with laser scanning sensor for different canopy volumes and foliage densities
[121]	2012	Infrared	Identify the green targets using the infrared detection technology by the automatic detection device
[122]	2012	Ultrasonic	Attain uniform spray deposition and coverage for nursery liner applications by variable-rate Ultrasonic sensor system
[123]	2012	Multi-sensors	Discussed the development status and challenges of the detection systems of smart sprayers
[124]	2013	LIDAR	Examine the relation of tree canopy and the leaf area density for orchards by LIDAR 3D Measurement System
[125]	2013	Ultrasonic	Reduction of agrochemicals used as Plant Protection Products and achieving suitable deposition on the canopy
[126]	2016	Machine Vision	Automatic detection of diseases by Sensor-based variable-rate sprayer
[127]	2016	LIDAR	Integration of sensors and algorithms in sprayer and real-time regulation of the sprayer flow according to the plant structures
[128]	2016	Machine Vision	Development of Real-time variable-rate sprayer with a camera sensor for precision herbicide application
[129]	2017	Ultrasonic	Development of Model for Planar Orchard target-oriented variable-rate spraying
[130]	2018	LIDAR	Reduce pesticide to match the canopy characteristics by variable-rate orchard sprayer based on laser scanning sensor

four different approaches of fusion as described by Faceli et al. [108]:

- Complementary Fusion: In this type of fusion, different sensors measure different parts or features of the location, and then they are fused for the useful information.
- Competitive Fusion: In this case, all sensors are computing the same part or aspects of the environment to make the system more reliable.
- Cooperative Fusion: Fusion of this type allows the acquisition of new data. New information can be acquired by fusing the data from different sensors, which cannot be acquired by any of the sensors alone.
- Independent Fusion: Fusion of data from unrelated sensors through the storage of their information into a common data structures.

Different algorithms and methods are used in the fusion method, for example, Weight Medium, Kalman Filter, Classic and Bayesian Application, Dampster-Schaefer and Artificial Neural Network. The fusion of multimode sensors has been widely used to minimize the characteristic disadvantages and add the advantages of several types of sensors [109–111]. Data fusion methods have been commonly used in multisensory environments to join and collect data from various sensors. The objective of data fusion in a multidimensional environment is to achieve a minor probability of recognition error and greater consistency by using data from highly distributed sources [112]. There are differences between the laser scanner data and the camera images. The 2D laser scanner creates a unique horizontal scan of the environment, while the camera provides detailed information about the situation. The laser scanner provides distance and vector information, while the camera offers intensity and color information. Both types of data have regular performance. For example, various angles and boundaries correspond to an unexpected change in the range of laser scan data and a sudden change in the concentration of the image [113].

A multi-sensory system developed to increase the environmental awareness of an agricultural vehicle working in farm fields. Various sensor technologies are available, namely stereovision, LIDAR, ultrasonic and thermography. To combine them, new methods are proposed for automatically detecting obstacles and crossing unfeasible areas. The experimental results acquired in the agricultural perspective are presented to demonstrate the efficiency of the proposed systems. Sorting results were better with combined sensors than with individual sensors [114]. Experimental platforms demonstrated the advantage of using sensory fusion. A smart decision system on multisensory fusion based on a neural network has been developed to forecast the occurrence of diseases or stress in plants to treat them in a spatially variable way. Many strong sensor platforms have been installed and calibrated that combine optical sensing, GPS (Geostationary positioning system) and data processing unit. Field tests were performed to optimize the functioning of the multisensory disease detection device [115]. Shalal et al. [116,117] introduced a tree canopy recognition algorithm based on the camera and laser scanner data fusion. A laser scanner senses the edge points. Determines the thickness of tree canopy and Nordic objects when using camera images. Check the color and parallel edges of the tree canopy and other objects. Data integration of these sensors has been found to increase plant detection so that a laser scanner can be reliably supplied the range, angles and width of tree canopies. However, the visual method can differentiate between tree canopies and other objects that grow with different characteristics.

The fusion of the vision system and laser sensors increases the detection of vegetation since the laser scanner can deliver precise distances, angles and sizes of trees and objects. However, the vision system can differentiate plants from other objects (Table 2).

9. Use case implementation of sensor technology

In this review paper, a machine vision spraying system is taken as a research example for a better understanding of the development procedure of the sensor System. The crop bed detection in the greenhouse bed operated by the spraying system is the primary function of this study, which is mainly used to detect plants and Spray point location section for targeted spraying.

The hardware configuration of the automatic spraying platform system is divided into three major parts, the image vision unit, electric control unit, and spraying unit. The image vision unit is consists of a Raspberry Pi (Raspberry Pi 3 B +) single-board computer combined with a camera. The image camera used in this spraying platform is a simple image camera, mounted on the spray unit.

The vision unit of this spraying platform uses a Raspberry Pi single-board computer to receive the camera's images of plants. Image, used to identify the location of the planting medium and the position of the plant, using an open-source computer vision library (Open CV) [94].

The image spray position signal is transmitted to the spray controller unit. The Arduino MEGA 2560 single-chip microcontroller was used as the spray controller. The controller is responsible for judging the image of the plants captured by the image vision unit to determine the appropriate spraying point position and start spraying stage as shown flowchart in Fig. 8. When the spray stage is running, Arduino first collects the image data captured by the image camera from the image vision unit. Calculate the optimal spraying point through the Raspberry Pi single-board computer, and send the spraying judgment parameters to the Arduino terminal, and converts them into signals for electric components for spraying operations, and then wait for Arduino to finish spraying. The spray-

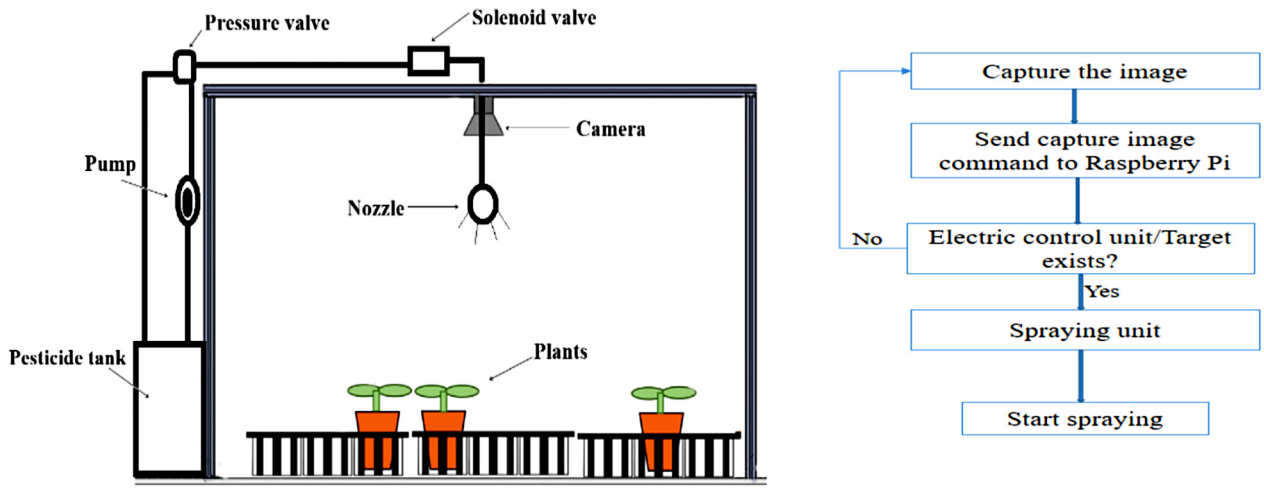

Fig. 8. Schematic diagram of machine vision spraying system and spraying control flow chart.

ing stage adopts the, and it will automatically power off when the fixed water pressure is reached.

In recent years, several Machine vision-based techniques have been introduced for imaging-based targeted spaying applications. These techniques are beneficial for saving the pesticide, which should be exactly deposited on the targets and reduction in pesticide usage [90,101]. The proposed system is successfully applied and tested the overall spraying success rate of the machine vision Sensor-based automatic spraying system in the greenhouse plants. The automatic spraying system can visually identify the plant position for targeted spraying. Calculate the spray point, and the valve opening time is used to control the spraying volume. In the function of the machine vision recognition system, the best Spraying position can be successfully identified and calculated when the plants' image recognition experiments are randomly placed in different plant postures. The average spraying success rate was 94.74 %. The proposed system is successfully applied to the identification of plants. Our main study motivation is to reduce the amount of pesticides by machine vision targeted spraying in green house for small plants. The proposed system will helps the researchers for the developing of variable-rate spraying systems based on real-time sensors technologies (Table 3).

10. Discussion

The intelligent sprayer prototype with smart functions for nurseries and orchards developed in this research has several advantages over conventional chemical application systems. Variable-rate control technologies to adjust the amount of spray output based on the need of target canopies. It can not only reduce the chemical cost for the growers but also reduce the risk of potential environmental contamination due to a significant reduction in the chemicals that miss the targets. It achieves fully automatic variable-rate spray tasks without human involvement during spray applications.

For nearly three decades, various scientists and engineers have proposed and developed multiple procedures and methods for detecting tree cover. The development of new alternative and environmentally friendly spraying techniques at the variable rate has only started in the past four decades. The goal was to use pesticides as low as possible and application of a sprayer with variable flow only when necessary, with minimal environmental damage. Different techniques and approaches for tree cover recognition have been recommended and established by scientists [131]. Detailed research shows that there is a suitable structure. One of the severe plant safety problems is the variable-rate sprayer. A modification sensor is an appropriate tool for determining the characteristics of the intermediate canopy. Allow the sprayer dose to be managed more precisely in variable proportions. Some researchers have developed various kinds of sensors, such as infrared sensors, ultrasonic sensors, laser sensors, etc. They suggested that it could be used effectively to characterize the exact canopy. Table 4. Shows the basic principles and characteristics of the real-time sensors technologies.

The analysis of sensory systems for the characteristics of the electronic canopy shows that infrared sensors and ultrasonic sensors are among the oldest and simplest sensors, but remain an appropriate tool for defining the properties of the canopies, such as large spaces between the ends of the rows and trees well-spaced. An ultrasonic sensor is a suitable tool for determining the characteristics of the canopy, while a LIDAR sensor delivers more precise and comprehensive information on the canopy. It is more appropriate to use an infrared sensor on variable rate sprayers than others. Also, infrared and ultrasonic sensor transmitters can be used to measure tree canopy density if equipped with the appropriate software. Therefore, we consider that this form of the sensor will remain in the sprayers in the future, as this could simplify operator intervention.

The most advanced feature of the machine vision system is a very attractive sound for tree canopy detection, as this method captures a huge image of an orchard in less time. However, the computer produced 3D digital area model of the farm does not yet provide enough functionality to cover the width, altitude and number of leaves to estimate the leaf area index. The optical sensing methods for the electronic characteristics of the canopy, with a LIDAR sensor, offer the maximum precise and comprehensive data on the tree canopies. If supported by the appropriate software, a LIDAR-based signal can provide the ideal device for generating 3D space with low costs. For all these aims, LIDAR will characterize an essential sensor in the future progress of spraying prototypes.

Despite its low cost, the usage of infrared sensors is in the early phases of the investigation. All sensors studied provide useful information to determine the spraying factors. In terms of plant protection, they offer suitable biological efficiency during application.

Table 3
Machine vision spraying system spraying success rate results.

Sr. no	Successful identification (Plants)	Identification failed (Plants)	Recognition success rate	Failure
1	13	1	92 %	Spray position failure
2	14	0	100 %	No
3	13	1	92 %	Spray position failure
4	12	2	85.7 %	Spray position failure
5	14	0	100 %	No
6	14	0	100 %	No
7	12	2	85.7 %	Spray position failure
8	14	0	100 %	No
9	13	1	92 %	Spray position failure
10	14	0	100 %	No

Table 4
Basic concept of sensors and their key advantages and disadvantages.

Sensor	Basic Theory	Advantages	Disadvantages
Infrared sensors	Detection of infrared light radiation emitted or reflected from plants.	- Low cost - daytime and nighttime reliable - low power requirements - Little impact of temperature and humidity on target detection.	- Limited range - Cannot detect the canopy characteristic information - Deficient spatial resolution for use in agriculture. - Light intensity has apparent effects on detecting
Ultrasonic sensors	Evaluate the distance of an object by calculating sending and receiving time of the sound wave	- Easily interface with microcontroller - Easy to use - Not affected by the colour of objects - low cost	- Susceptible to background noise - Limited detection range - Many sensors used for common agricultural scene - Influenced by weather
LIDAR sensors	Measures the distance to object using the time of flight of laser beams	- Higher accuracy - Fast Working - Used in any lighting conditions - The structure of trees describe with accuracy	- Poor performance in edge detection - Have problems under adverse weather conditions - High cost
Machine vision	Provide imaging-based automatic investigation and analysis for automatic assessment, controlling, and machine guidance	- Robust enough for open-field applications - Gives a large quantity of information - Provides an accurate 3D image of plants canopies - Gives high precision structure information of trees with	- Extensive computation requirement - Susceptible to direct sunlight - High cost - Need appropriate calibration and recording procedures
Sensor Fusion	The process of integrating multiple sensors information to achieve better reliable, precise, and valuable data than that delivered by any single sensor	- Makes information more significant and precise than a single sensor - Deliver a improved estimation of precision over a wide choice of working situations - Reduce the errors	- Sensor fusion systems are high cost - Sensor fusion required heavy processing, and increased system complexity and require faster processors.

11. Conclusion and future prospects

Variable-rate chemical application based on target detection is significant to meet the environmental, commercial and protection measures of good agricultural practices. The researchers tried to invention better management methods for applying discreet and sustainable decision models but again unsuccessful in the case of irregular trees rising in a fruit wall system. Many researchers have developed systems to adapt the degree of application to changes in the parameters of the tree canopy structure, using different assessment models and many sensitive technologies. Vision and laser sensors, a variable degree of spraying in orchards, are still being tested, however, the alternative spraying method can only be developed in the case of large orchards with fully automatic trailers and self-propelled platforms. In general, computer vision with appropriate image processing algorithms is a very promising tool for detecting plants and weeds at the right time, providing valuable sensitive information. While there are challenges to be faced, there is hope for the future as many researchers are focusing their efforts on this task. Most of the studies are conducted under ideal states. Due to the uncontrolled environment and the limited use of sensors in agricultural production, there are considerable technical difficulties in automation, such as lighting conditions are one of the main tasks for the application of machine vision and spectral investigation. Especially for the machine vision, this is a significant feature of crop recognition and discrimination. Many sensors can be used to measure each target. Susceptible data to limit each sensor can be integrated to increase detection precision or develop an efficient spraying method. All of these techniques can characterize targets with acceptable precision.

In the future, the advancement and progress of innovative sensors dedicated to geometry. The classification of tree plants will allow important and much-needed improvement for optimization and increased production and quality, as well as the use of variable-rate sprayers in food production by educating training organizations. It should be noted that the benefits of variable spray-

ing affect millions of hectares of crops and therefore has a direct impact on our society and our environment. It is therefore essential to continue making great development efforts. More precise, robust and convenient systems able to measure the geometric structural characteristics of the orchards that are consistently providing the progress of various fields and precise agriculture. But some technological solutions are still needed. The first is to improve research systems to develop software and increase the speed of computation and for the subsequent processing phase's decision making. Among these, it's essential to develop low-cost sensors and control systems that assist large-scale applications. 3D imaging technology is improving, so the potential of this technology is enormous. Reasons such as the reduction of the workforce, the scarcity of natural resources and the consumer demand for quality products create a need for automation in agriculture. Since the Real-Time Sensor system is a crucial technology for variable rate target spraying, more implementations are yet to come.

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CRediT authorship contribution statement

Irfan Abbas: Data curation, Formal analysis, Funding acquisition, Investigation, Writing - original draft. **Jizhan Liu:** Supervision, Conceptualization, Methodology, Writing - review & editing. **Muhammad Faheem:** Project administration, Resources. **Rana Shahzad Noor:** Software, Validation, Visualization. **Sher Ali Shaikh:** Investigation, Writing - review & editing. **Kashif Ali Solangi:** Writing - review & editing. **Syed Mudassir Raza:** Formal analysis, Writing - review & editing.

Declaration of Competing Interest

The authors declared no conflict of interest.

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